

Computer Science & IT

Database Management System



Transaction & concurrency control

Lecture No. 01



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Recap of Previous Lecture



Topic

Practice questions

Topic

Domain relational calculus (DRC)

Topics to be Covered



- ✓ Topic Transaction
- ✓ Topic ACID properties
- ✓ Topic Atomicity
- ✓ Topic Durability





Topic : Transaction



A sequence of logically related operation which is used to perform a particular task is called a transaction

Read(A) :-
Read of data item 'A'

This operation is used to read the value of dataitem 'A' from the database (ie. from the disk) into Main Mem.

Write(B) :-

Update the new value of dataitem B into the database, Writ. value of B present in Main memory

let initially , amount in account 'A' = 1000/-

and amount in account 'B' = 0/-

Write the transaction to transfer Rs 500/- from account A to account B.

Transaction

Read (A); : Because of this opⁿ a copy of data item 'A' will be created in M.M.

A = A - 500; // This updation will be performed only in M.M.

Write (A);

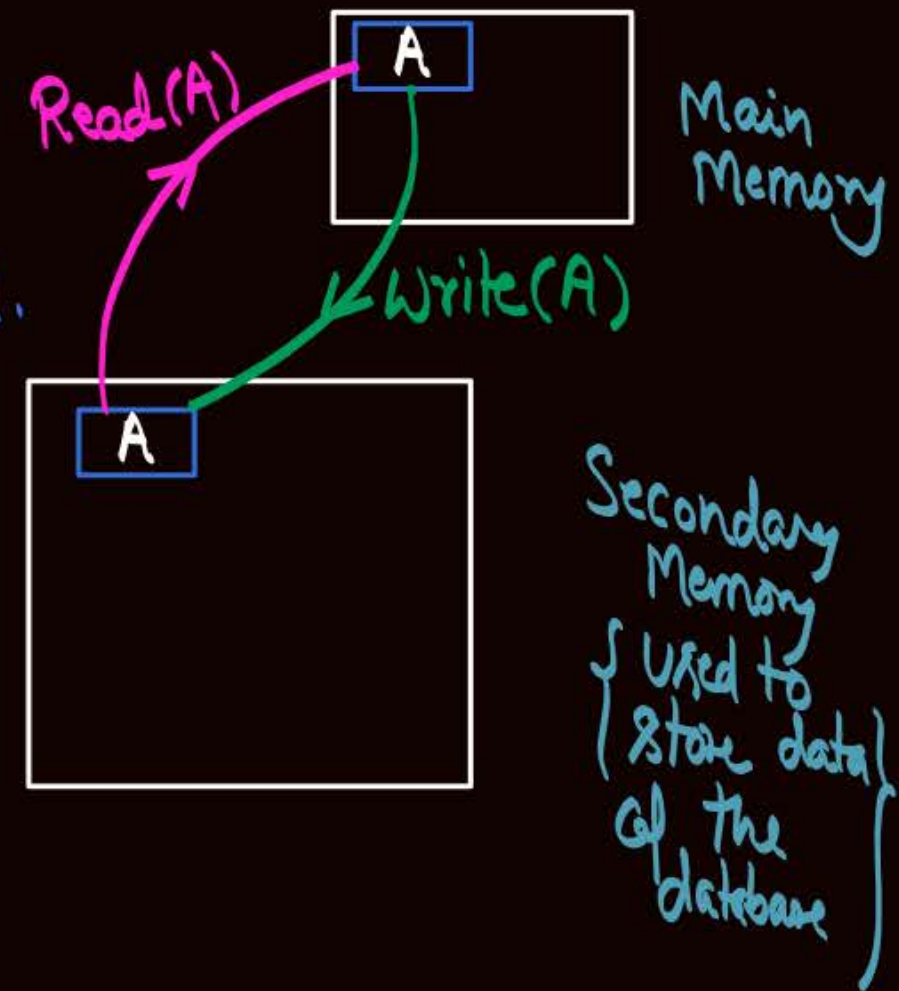
Read (B);

B = B + 500;

Write (B);

Commit; { Transaction Committed means transaction executed successfully from user point of view }

These two operations are not important from database Point of view.





Topic : Blind write



Transaction(T)

R(A)

A = A - 10;

W(A)

W(B);

← It is not the blind write opⁿ, because T has already read the dataitem 'A' before it updates 'A'

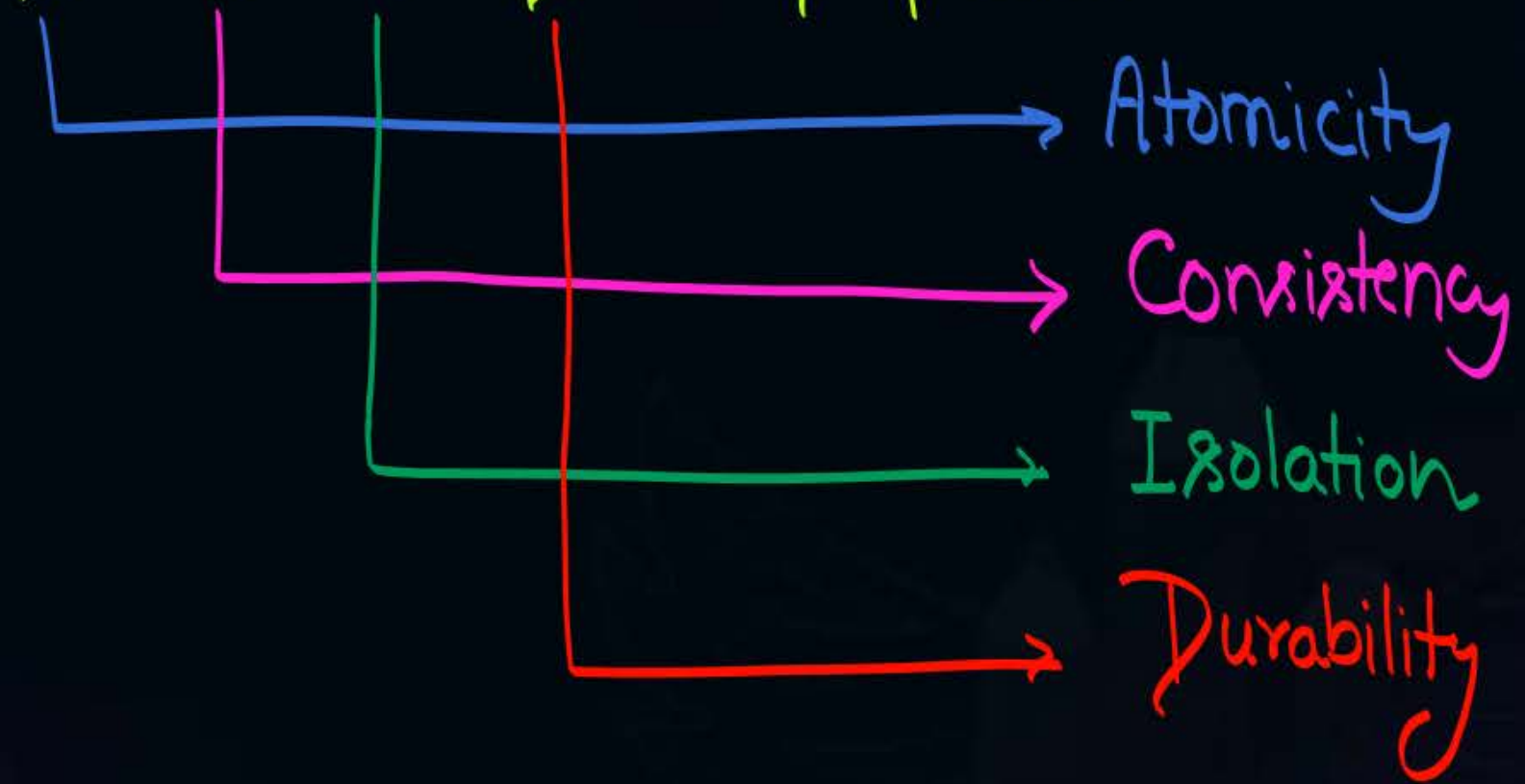
{ Transaction T did not read the data-item 'B', and it is trying to update the dataitem 'B' without reading the value of dataitem B from the database. Such write opⁿ are called "Blind write" opⁿ



Topic : ACID properties



For the integrity of the database the transactions of database must satisfy A C I D Properties.





Topic : Atomicity



Atomicity states that either execute all operations of a transaction or execute none of them.

There are two problems:-

① All the operations of a transaction can not be executed at once, they will get executed in a sequence one after another.

② If we start executing the operations of a transaction then we can not guarantee that the transaction will not fail in between { i.e., transaction may fail after executing some of the operations }

let initially , amount in account 'A' = 1000/-

and amount in account 'B' = 0/-

Write the transaction to transfer Rs 500/- from account A to account B.

Transaction

Read (A);

A = A - 500;

Write (A);

Read (B);

B = B + 500;

Write (B);

Commit; { Transaction Committed means transaction executed successfully from user point of view }

If we start executing the operations of this transaction and if transaction fails anywhere before successfully executing the Commit opn then atomicity is violated.

Reason of failure may be anything.

eg

- (i) Power failure
- (ii) Software Crash
- (iii) Disk Crash
- (iv) Any natural calamity
- (v) OS or DBMS may kill a transaction if required.

etc.,

⋮

Note :- →

To ensure atomicity, if a transaction fails before executing its commit operation successfully, then we must "undo" the changes performed by that transaction before its failure point.

{ If we can do so, the behaviour will be similar to the situation when none of the operation of that transaction is executed }

To ensure atomicity "Recovery Management Component" is used by DBMS.



Topic : Recovery management component

- * Recovery management Component is responsible for "undo" and "redo" operation

"Redo" opn will be performed w.r.t. Committed transaction.

"undo" opn will be performed w.r.t. failed transactions.

- * The process of "undoing" the operations performed by a failed transaction is called "rollback" of that transaction
- * For the purpose of "undo" and "redo" operation, "transaction log" is maintained by recovery management Component.



Topic : Transaction log

- * "Transaction log" is used to record the activities performed by the transactions.

let initially, amount in account 'A' = 1000/-
and amount in account 'B' = 0/-

Write the transaction to transfer Rs 500/- from account A to account B.

Transaction

Read (A);

$A = A - 500;$

Write (A);

Read (B);

$B = B + 500;$

Write (B);

Commit;

Fails



"Transaction log" W.r.t transaction to transfer
Rs 500/- from account A to account B

Transaction log

A.old = 1000

A.new = 500

transaction
Fails

In order to rollback the transaction, we must update the values in the database by the old values present in the transaction log for those data items.



Topic : Transaction log

* "Transaction log" is used to record the activities performed by the transactions.

→ We can recover from a failure only if transaction log is available,

o Transaction log is maintained by recovery management component into the secondary mem.

* Either we can maintain a separate transaction log
(or) we can maintain a unified transaction log
for each ongoing transaction.

Consider 3 transactions, T₁, T₂, T₃

(T, X, P, Q)

transaction id data item old value new value

T₁: Start;
(T₁, A, 50, 100)
(T₁, B, 70, 50)
T₂ start;
(T₂, C, 100, 200)
Check point;
(T₁, D, 200, 100)
(T₂, E, 50, 500)
T₂ Commit;
T₃ Start;
(T₃, F, 100, 550)

We don't need to "redo" the opⁿ performed by T₂ before last "Check point".

The opⁿ before the Check point have been reflected into DB.

We will redo this opⁿ

Because T₂ Committed

Failure

T₁ & T₃ were Ongoing opⁿ When failure occurred
∴ We must "undo" all the operations Performed by T₁ & T₃.



Topic : Durability



- ★ Changes performed by committed transaction must persist in the database even in the case of failure.



2 mins Summary



✓ **Topic**

Transaction

✓ **Topic**

ACID properties

✓ **Topic**

Atomicity

✓ **Topic**

Durability

THANK - YOU